Quadratic-Time Balanced Minimum Evolution Discrete Gradients, Column-Tiled SIMT Parallelism, and Topological Tree Optimization in Phylo2Vec Space

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Abstract

Balanced Minimum Evolution (BME) is one of the most statistically consistent and widely used distance-based criteria for phylogenetic tree inference, powering standard tools such as FastME. Classical BME optimizers represent trees as pointer-linked graph structures and perform sequential Subtree Prune and Regraft (SPR) hill-climbing, applying a single topological move per $O(N^2)$ sweep. In contrast, the Phylo2Vec representation bijectively encodes any rooted binary tree on N leaves as an integer vector $v \in \mathbb{Z}^{N-1}$ satisfying $0 \le v_i \le 2i$. In this paper, we present a complete mathematical formulation and high-performance software architecture for exact $O(N^2)$ BME discrete gradient computation and topological optimization directly in Phylo2Vec vector space. First, we derive the complete $(N-1) \times (2N-3)$ discrete gradient matrix $\nabla_{\text{disc}} L(v)$ over all valid coordinate values $0 \le j \le 2i$, paired with an $O(N)$ top-down cherry-first gauge transformation and an $O(N)$ inverse tree-to-vector bijection. Second, we introduce a node-disjoint multi-coordinate (K-SPR) batch optimizer with dual-mode gauge rotation and stochastic 2-SPR basin hopping that applies multiple non-interfering SPR moves per $O(N^2)$ gradient evaluation. Third, to scale to massive trees on multi-core CPUs and GPUs, we prove four architectural theorems: (i) the *Column-Independence Theorem* for lockstep SIMT clade-distance dynamic programming, (ii) the $O(N \log N)$ *Stackless Ancestor-Walk Top-Down Theorem*, (iii) the *Heavy-First* $\lceil \log_2(2N) \rceil$*-Slot Register-Stack Theorem* with $O(1)$ incremental destination mapping, and (iv) the *Complete Column-Autonomy Theorem* for $4\times$ memory-reduced column tiling. Across 120 benchmark datasets ($N = 20 \ldots 200$), our optimizer achieves a 100% win/tie rate in BME tree length against FastME while requiring up to $3.16\times$ fewer $O(N^2)$ evaluations and running up to $2.97\times$ faster on a single core. At large scale ($N = 100 \ldots 4{,}000$), our column-parallel and GPU-accelerated kernels achieve up to a $6.79\times$ speedup on an Apple M5 Metal GPU and enable tree optimization up to $N = 350{,}000$–$400{,}000$ taxa on $8\times$ NVIDIA H100 and $8\times$ NVIDIA RTX PRO 6000 nodes.

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1 Introduction

Under the Balanced Minimum Evolution (BME) framework introduced by Pauplin [1] and formalized by Desper and Gascuel [2], the length $L(T)$ of an unrooted binary phylogenetic tree T with N leaves and pairwise leaf distance matrix $D \in \mathbb{R}^{N\times N}$ is defined as:

$$L(T) = \sum_{x=0}^{N-1} \sum_{\substack{y=0 \\ y\neq x}}^{N-1} D_{xy}\, 2^{-d_T(x,y)}, \tag{1}$$

where $d_T(x, y)$ denotes the number of internal degree-3 nodes on the unique path between leaf x and leaf y in T (so that the number of edges on the path is $d_T(x, y) + 1$). Desper and Gascuel [2] and Lefort et al. [3] (FastME 2.0) showed that all $O(N^2)$ Subtree Prune and Regraft (SPR) moves on T can be evaluated in $O(N^2)$ total time via a two-pass recursive tree traversal that maintains balanced clade-to-clade average distances.

While FastME's $O(N^2)$ per-sweep complexity is asymptotically optimal for evaluating all SPR moves of a fixed tree, pointer-based graph representations impose three fundamental computational limitations:

1. **Single-Move Sequential Trajectory ($\Theta(N^3)$ Scaling):** FastME identifies the single best SPR move across all $O(N^2)$ candidates, mutates the pointer graph for that one move, and recomputes the entire $(2N - 2) \times (2N - 2)$ balanced averages table from scratch on the next iteration. Because optimizing an N-leaf tree requires $\Theta(N)$ SPR moves, total optimization time scales as $\Theta(N^3)$ sequential sweeps.

2. **Irregular Pointer Chasing:** Representing trees via heap-allocated node and edge structs prevents SIMD vectorization and SIMT GPU execution, causing severe CPU cache and TLB misses once the clade-distance table exceeds L2/L3 cache ($N \geq 1{,}500$).

3. **Lack of Algebraic Coordinate Gradients:** Pointer graphs do not expose a fixed-dimensional coordinate domain for discrete gradient descent, batch coordinate updates, or variational phylogenetic inference.

Recently, Penn et al. [4] introduced **Phylo2Vec**, a bijective representation that maps any rooted binary phylogenetic tree on N labeled leaves $\{0, 1, \ldots, N - 1\}$ to an integer vector $v = (v_0, v_1, \ldots, v_{N-2}) \in \mathcal{V}$, where:

$$\mathcal{V} = \left\{ (v_0, v_1, \ldots, v_{N-2}) \in \mathbb{Z}^{N-1} \;\middle|\; 0 \leq v_i \leq 2i \quad \forall\, i \in \{0, \ldots, N-2\} \right\}, \qquad |\mathcal{V}| = (2N-3)!!. \tag{2}$$

Perturbing a single coordinate $v_i \leftarrow j$ ($j \in \{0, \ldots, 2i\}$) corresponds to a structured SPR move in which the subtree attached at step i is pruned and regrafted onto the edge indexed by j in the prefix tree on leaves $\{0, \ldots, i\}$.

1.1 Contributions and Architecture Overview

In this paper, we describe a complete, self-contained mathematical and software framework for BME discrete gradient computation and topological tree optimization in Phylo2Vec space:

1. **Full $0 \leq j \leq 2i$ Discrete Gradient in $O(N^2)$ Time (Section 2):** For any tree in cherry-first canonical gauge ($0 \leq v_i \leq i$), we compute the complete $(N - 1) \times (2N - 3)$ discrete gradient matrix $\Delta_{i,j} = L(v^{(i\leftarrow j)}) - L(v)$ across all $2i + 1$ valid regrafting coordinates $0 \leq j \leq 2i$—covering both leaf-incident pendant edges ($0 \leq j \leq i$), internal clade edges ($i + 1 \leq j \leq 2i - 2$), and root-incident edges ($j \in \{2i - 1, 2i\}$).

2. **Exact $O(N)$ Cherry-First Gauge Transformation and Inverse Bijection (Section 2.2):** We prove that every binary tree can be relabeled into the cherry-first submanifold $\mathcal{V}_{\text{cf}} = \{v \in \mathcal{V} : 0 \le v_i \le i\}$ in $O(N)$ time via a single top-down Euler tour, and we provide an exact $O(N)$ inverse bijection (`encode_rooted_to_v_c` and `encode_unrooted_to_v_c`) that recovers $v \in \mathcal{V}$ from any rooted or unrooted tree topology.

3. **Node-Disjoint Multi-Coordinate (K-SPR) Batch Optimizer (Section 3):** We design a hybrid optimizer that extracts multiple mutually node-disjoint improving coordinate moves from a single discrete gradient evaluation, applying K non-interfering SPR moves simultaneously with exact line-search verification, dual-mode gauge rotation, and 2-SPR basin hopping.

4. **SIMT GPU, Multi-Core CPU, and Column-Tiled Algorithms (Section 4):** We prove four theorems that transform the tree-based recurrences into lockstep vector/matrix operations: the *Column-Independence Theorem*, the $O(N \log N)$ *Stackless Ancestor-Walk Theorem*, the *Heavy-First* $\lceil \log_2(2N) \rceil$*-Slot Register-Stack Theorem* with $O(1)$ incremental destination mapping, and the *Complete Column-Autonomy Theorem* for $4\times$ memory-reduced column tiling.

2 Mathematical Formulation of the Phylo2Vec Discrete Gradient

2.1 Phylo2Vec Tree Decoding and Unrooted Representation

Given a vector $v \in \mathcal{V}$ in the cherry-first canonical domain ($0 \le v_i \le i$ for all $i \in \{0, \dots, N-2\}$), the corresponding rooted binary tree T_{root} with $2N-1$ nodes (leaves $0 \dots N-1$ and internal nodes $N \dots 2N-2$) is decoded in $O(N)$ time in reverse coordinate order $k = 0, \dots, N-2$ (where $i = N-2-k$):

$$\text{pairs}[k] = (p_1, p_2) = (v_i,\ i+1), \qquad m = N + k. \tag{3}$$

Using a path-compressed union-find array parents_uf$[0 \dots 2N-2]$ initialized to the identity, each step k finds the current subtree roots $r_1 = \text{find}(p_1)$ and $r_2 = \text{find}(p_2)$, creates internal node $m = N + k$ with children $c_1(m) = r_2$ and $c_2(m) = r_1$, and sets parents_uf$[r_1] =$ parents_uf$[r_2] = m$.

Because the BME tree length $L(T)$ in Eq. (1) is invariant to the placement of the root on the top edge, we collapse the rooted root node $2N-2$ (whose children are $R = 2N-3$ and $c_{\text{other}} = \min(c_1(2N-2), c_2(2N-2))$) into an unrooted tree T on $M = 2N-2$ nodes $\{0, 1, \dots, 2N-3\}$ rooted at the degree-3 trifurcating node $R = 2N-3$ with three children:

$$R_{c_0} = c_1(R), \qquad R_{c_1} = c_2(R), \qquad R_{c_2} = c_{\text{other}}. \tag{4}$$

Every internal node $u \in \{N, \dots, R-1\}$ has two downward children $c_1(u), c_2(u)$ and one upward parent parent_u$[u]$, forming a 3-regular adjacency table adj_u$[u, 0..2] = (c_1(u), c_2(u), \text{parent_u}[u])$, while adj_u$[R, 0..2] = (R_{c_0}, R_{c_1}, R_{c_2})$. We compute subtree node counts sub_size$[u]$ bottom-up and assign contiguous Depth-First Search (DFS) entry times tin$[u] \in \{0, \dots, 2N-3\}$ top-down:

$$\begin{aligned} &\text{tin}[R] = 0, \quad \text{tin}[R_{c_0}] = 1, \quad \text{tin}[R_{c_1}] = 1 + \text{sub_size}[R_{c_0}], \quad \text{tin}[R_{c_2}] = \text{tin}[R_{c_1}] + \text{sub_size}[R_{c_1}], \\ &\text{tin}[c_1(u)] = \text{tin}[u] + 1, \quad \text{tin}[c_2(u)] = \text{tin}[u] + 1 + \text{sub_size}[c_1(u)] \qquad (u = R-1 \text{ down to } N). \end{aligned} \tag{5}$$

By construction, a node v lies in the strict downward subtree $T_u \setminus \{u\}$ of internal node u if and only if $\text{tin}[u] < \text{tin}[v] < \text{tin}[u] + \text{sub_size}[u]$, and for $v \in T_u \setminus \{u\}$, v lies in $T_{c_1(u)}$ if and only if $\text{tin}[v] < \text{tin}[c_2(u)]$.

2.2 Exact $O(N)$ Cherry-First Gauge Transformation and Inverse Bijection

While general Phylo2Vec vectors satisfy $0 \leq v_i \leq 2i$, the bottom-up union-find decoding above requires the *cherry-first canonical gauge* $0 \leq v_i \leq i$ (in which every merge occurs on a leaf pendant edge rather than an internal ancestral edge). We establish that any binary tree can be transformed into cherry-first gauge in $O(N)$ time via a leaf permutation $\pi \in S_N$, and conversely any rooted or unrooted tree can be encoded back into $v \in \mathcal{V}$ ($0 \leq v_i \leq 2i$) in $O(N)$ time.

Theorem 2.1 ($O(N)$ Cherry-First Gauge Transformation and Clade-Flip Duality)**.** *Let T be any rooted binary tree on N leaves decoded from $v \in \mathcal{V}$ ($0 \leq v_i \leq 2i$). For every node $u \in \{0, \ldots, 2N-2\}$, define the minimum leaf in u's clade bottom-up by $m(x) = x$ for $x < N$ and $m(u) = \min(m(c_1(u)), m(c_2(u)))$ for $u \geq N$. Fix a clade orientation flip_mode $\in \{0, 1\}$, and propagate leaf labels $\ell(u) \in \{0, \ldots, N-1\}$ top-down from the root $\ell(2N-2) = 0$:*

- *If flip_mode $= 0$, order the children (a, b) of u so that $m(a) < m(b)$; if flip_mode $= 1$, order (a, b) so that $m(a) > m(b)$.*
- *Set $\ell(a) = \ell(u)$ and $\ell(b) = \ell(u) +$ leaf_count(a), where leaf_count(a) is the number of leaves in T_a.*
- *At each internal node u, record the cherry-first coordinate at index $i = \ell(b) - 1$ as $v_i^{cf} = \ell(a) < \ell(b) = i + 1$.*

Then $\pi(k) = \ell^{-1}(k)$ defines a bijection on $\{0, \ldots, N-1\}$ such that $v^{cf} \in \mathcal{V}_{cf}$ ($0 \leq v_i^{cf} \leq i$ for all i) encodes the exact same tree T under the permuted distance matrix $D_\pi[a, b] = D[\pi(a), \pi(b)]$, with $L(v^{cf}; D_\pi) = L(v; D)$.

Proof. At every internal node u, the subtree T_a contains leaf_count(a) leaves assigned to the contiguous integer interval $[\ell(u),\ \ell(u)+\text{leaf_count}(a)-1]$, with minimum leaf $\ell(a) = \ell(u)$, while T_b is assigned to $[\ell(b),\ \ell(b)+\text{leaf_count}(b)-1]$ with minimum leaf $\ell(b) = \ell(u)+\text{leaf_count}(a) > \ell(a)$. When T_a and T_b merge in bottom-up decreasing order of $\ell(b)$, all internal merges within T_a and T_b involving larger minimum leaves $\ell > \ell(b)$ have already completed. Thus T_a and T_b are represented by their minimum leaves $\ell(a)$ and $\ell(b) = i + 1$, so the merge is encoded by the leaf pair $(v_i^{\text{cf}},\ i+1) = (\ell(a),\ \ell(b))$ with $v_i^{\text{cf}} = \ell(a) \leq i$. □

Theorem 2.2 ($O(N)$ Exact Inverse Bijection from Trees to $\mathcal{V}$)**.** *Given any rooted binary tree T on leaves $\{0, \ldots, N-1\}$ (or any unrooted binary tree T rooted at leaf $N-1$'s incident internal node), let each internal merge $m \in \{N, \ldots, 2N-2\}$ combine child clades with minimum leaves $l_{\min}(m) < l_{\max}(m)$. Then $l_{\max}(m)$ takes each value in $\{1, \ldots, N-1\}$ exactly once. Ordering the merges in increasing order of $l_{\max}$ (pair_lmax_order$[i]$ for $i = 0, \ldots, N-2$), let $u_{attach}(i)$ denote the child of merge pair_lmax_order$[i]$ having minimum leaf $l_{\min}$. If $u_{attach}(i) < N$ is a leaf, then $v_i = l_{\min}$; if $u_{attach}(i) \geq N$ is an internal clade, then:*

$$v_i = (2i + 1) - \mathit{rank}(u_{attach}(i)), \tag{6}$$

where $\mathit{rank}(u_{attach}(i)) \in \{1, \ldots, i\}$ is the number of prior merges $i' < i$ that occurred strictly after $u_{attach}(i)$ in bottom-up topological order (computable in $O(N)$ time).

2.3 Two-Pass $O(N^2)$ Balanced Clade Distance Matrix (`subdist`)

To evaluate all SPR moves in $O(N^2)$ time, we construct the symmetric $(2N-2) \times (2N-2)$ balanced clade distance matrix subdist$[u, v]$, which stores the directional balanced average distance between the subtrees induced by every pair of nodes (u, v) in the unrooted tree T:

1. **Leaf Initialization ($0 \leq x, y < N$):** subdist$[x, y] = D_{xy}$.

2. **Bottom-Up Clade Recurrence ($u = N, \ldots, R-1$):** For any node $v \notin T_u$, the balanced average distance from u's downward subtree T_u to v satisfies:

$$\text{subdist}[u, v] = \text{subdist}[v, u] = \frac{1}{2}\left(\text{subdist}[c_1(u), v] + \text{subdist}[c_2(u), v]\right). \tag{7}$$

3. **Trifurcating Root 3-Way Recurrence ($u = R = 2N-3$):** For each child R_{c_k} ($k \in \{0,1,2\}$) with complementary root children $R_{c_{k'}}, R_{c_{k''}}$ ($k', k'' \neq k$), every node $v \in T_{R_{c_k}}$ has:

$$\text{subdist}[R, v] = \text{subdist}[v, R] = \frac{1}{2}\left(\text{subdist}[R_{c_{k'}}, v] + \text{subdist}[R_{c_{k''}}, v]\right). \tag{8}$$

4. **Top-Down Complementary Clade Recurrence ($p = R-1$ down to N):** When $v \in T_p \setminus \{p\}$ lies inside the downward subtree of p, the clade at p oriented *away* from v consists of p's parent $g = \text{parent_u}[p]$ and the sibling child $\text{sib}_p(v) \in \{c_1(p), c_2(p)\}$ not containing v:

$$\text{subdist}[p, v] = \text{subdist}[v, p] = \frac{1}{2}\left(\text{subdist}[\text{sib}_p(v), v] + \text{subdist}[\text{parent_u}[p], v]\right). \tag{9}$$

5. **Diagonal and Local Neighbor Distances (`nd_u`):** Setting $\text{subdist}[u, u] = 0$ for all u, we cache for each internal node $u \in \{N, \ldots, R\}$ with neighbors $(n_0, n_1, n_2) = \text{adj_u}[u, 0..2]$ the three pairwise neighbor-clade distances:

$$\text{nd_u}[u, 0] = \text{subdist}[n_1, n_2], \quad \text{nd_u}[u, 1] = \text{subdist}[n_0, n_2], \quad \text{nd_u}[u, 2] = \text{subdist}[n_0, n_1]. \tag{10}$$

2.4 Half-Tree SPR Recurrence and the Full $0 \le j \le 2i$ Gradient Mapping

For each row $i \in \{1, \ldots, N-2\}$ (corresponding to leaf index $v_{\text{index}} = i+1$), the subtree pruned when coordinate v_i changes is $T_{\text{prune}} = \text{v_map}[v_{\text{index}}]$, which was attached to internal node $P = \text{parent_u}[T_{\text{prune}}]$ (or c_{other} if $T_{\text{prune}} = R$). Removing T_{prune} and contracting P leaves an edge between P's other two neighbors, other_0 and other_1, partitioning the remaining tree into two disjoint half-trees rooted at other_0 and other_1.

Traversing into the half-tree starting at $u_0 = \text{other}_0$ (with opposite anchor $\text{orig} = \text{other}_1$ and initial distance $d_{kC}^{(0)} = \text{subdist}[T_{\text{prune}}, \text{orig}]$), each step across an internal node u from predecessor C branches into the two outgoing neighbors b_0, b_1 of u with path-length weight $w_\ell = 2^{-(\ell+1)}$. Updating the path-adjusted cross-clade distance D_{AC} and pruned-clade distance d_{kC}:

$$\begin{aligned} D_{AC}(b_0) &= \text{subdist}[C, b_1]^{\text{before}} + \left(\text{subdist}[\text{orig}, b_1] - \text{subdist}[T_{\text{prune}}, b_1]\right) 2^{-(\ell+1)}, \\ \Delta L(u \to b_0) &= \Delta L(C \to u) + \frac{1}{4}\left(D_{AC}(b_0) + \text{subdist}[T_{\text{prune}}, b_0] - D_{AB}(u) - d_{kC}(u)\right), \\ d_{kC}(b_0) &= \frac{1}{2}\left(d_{kC}(u) + \text{subdist}[T_{\text{prune}}, b_1]\right), \end{aligned} \tag{11}$$

and symmetrically for b_1, we store $\Delta L(u \to b)$ in $\text{edge_delta}[c_{\text{end}}(u, b)]$, where $c_{\text{end}}(u, b) \in \{0, \ldots, 2N-3\}$ is the child endpoint of the edge (u, b) in the unrooted tree rooted at R. Running this recurrence on both $(\text{other}_0, \text{other}_1)$ and $(\text{other}_1, \text{other}_0)$ populates $\text{edge_delta}[0 \ldots 2N-3]$ in $O(N)$ time per row i.

Theorem 2.3 (Complete $0 \le j \le 2i$ Edge-to-Coordinate Mapping)**.** *For any cherry-first vector $v \in \mathcal{V}_{cf}$ ($v_i \le i$) and row $i \in \{1, \ldots, N-2\}$, the change in BME tree length $\Delta_{i,j} = L(v^{(i \leftarrow j)}) - L(v)$ for all $2i+1$ valid Phylo2Vec coordinates $j \in \{0, \ldots, 2i\}$ is given by:*

$$\Delta_{i,j} = \begin{cases} \textit{edge_delta}\big[\textit{dest_row}_i[j]\big], & 0 \le j \le i, \\ \textit{edge_delta}\big[(2N-2) - 2i + j\big], & i+1 \le j \le 2i-2, \\ \textit{edge_delta}\big[c_{\textit{other}}\big], & j \in \{2i-1,\ 2i\}, \end{cases} \tag{12}$$

where dest_row$_i[j]$ is the highest ancestor of leaf j formed strictly after step i in the bottom-up decoding order ($m \in \{N, \ldots, 2N-3-i\}$), and $c_{other} = \min(c_1(2N-2), c_2(2N-2))$ is the non-R child of the rooted root $2N-2$ representing the collapsed top edge of the unrooted tree.

Proof. At step i of the Phylo2Vec decoding process, $N-2-i$ internal merges have already occurred (corresponding to internal nodes $m \in \{N, \ldots, 2N-3-i\}$). For $0 \le j \le i$, regrafting onto coordinate j attaches T_{prune} to the current root of the clade containing leaf j, which is dest_row$_i[j]$. For $i+1 \le j \le 2i$, Phylo2Vec indexes internal clades in reverse order of their creation time: $m(i,j) = N + (2i-j)$. Because the unrooted tree collapses the top edge $(2N-2)$ into (R, c_{other}) and shifts internal node indices above the pruned merge by $N-2-i$, the $2i-2-i = i-2$ non-root internal edges map bijectively to $u = (2N-2) - 2i + j$ for $i+1 \le j \le 2i-2$, while the two rooted edges incident to the root of the prefix tree ($j = 2i-1$ and $j = 2i$) both correspond to the single unrooted edge indexed by c_{other}. □

3 Topological Optimization via Node-Disjoint K-SPR Search

Evaluating the full $(N-1) \times (2N-3)$ discrete gradient matrix $\Delta = \nabla_{\text{disc}} L(v)$ reveals the exact BME change of all $(N-1)^2$ single-coordinate SPR moves in a single $O(N^2)$ sweep. Rather than applying only a single coordinate update per sweep, we exploit the discrete gradient matrix to apply $K \ge 1$ non-interfering SPR moves simultaneously.

3.1 Node-Disjoint Path Corridors and Batch Line Search

Each candidate coordinate update $v_i \leftarrow j^*$ ($j^* = \arg\min_{0 \le j \le 2i} \Delta_{i,j}$ with $\Delta_{i,j^*} < -\epsilon$) corresponds to moving the pruned subtree $T_{\text{prune}}(i) = \text{v_map}[i+1]$ from its current attachment node $P(i) = \text{parent_u}[T_{\text{prune}}(i)]$ to target edge endpoint $U(i, j^*)$ along a unique path corridor $\mathcal{P}_T(P(i), U(i, j^*))$ in the unrooted tree T.

In the BME length formula (Eq. (1)), an SPR move along $\mathcal{P}_T(P(i), U(i, j^*))$ alters the topological path weights $2^{-d_T(x,y)}$ primarily for paths passing through $\mathcal{P}_T(P(i), U(i, j^*))$. Consequently, if a set of improving coordinate moves $\mathcal{M} = \{(i_1, j_1^*), \ldots, (i_K, j_K^*)\}$ satisfies:

1. **Vertex-Disjointness of Corridors:** $\mathcal{P}_T(P(i_a), U(i_a, j_a^*)) \cap \mathcal{P}_T(P(i_b), U(i_b, j_b^*)) = \emptyset$ for all $a \ne b$ (including source and target neighbors), and
2. **Coordinate Prefix Independence:** No move i_a modifies a subtree minimum-leaf index referenced by another move i_b,

then the K moves are topologically non-interfering and their combined BME improvement satisfies $\Delta L(\mathcal{M}) \approx \sum_{k=1}^{K} \Delta_{i_k, j_k^*} < 0$.

Our greedy batch selector (`optimize_phylo2vec_bme_c`) sorts all improving rows i in ascending order of Δ_{i,j^*} (most negative first), marks the tree nodes on $\mathcal{P}_T(P(i), U(i, j^*))$ in a boolean mask blocked$[0 \ldots 2N-3]$ as each move is accepted into $\mathcal{M}$, and then performs a fast backtracking line search over $K' \in \{|\mathcal{M}|, \lfloor |\mathcal{M}|/2 \rfloor, \ldots, 1\}$ using an $O(N)$ cherry-first gauge canonicalization and exact $O(N^2)$ BME evaluation, guaranteeing strict monotonic descent $L(v^{(t+1)}) < L(v^{(t)})$ on every step and exact single-move fallback when $K' = 1$.

3.2 Complete 4-Stage Phylo2Vec BME Optimization Pipeline

The complete topological optimizer (`optimize_phylo2vec_bme_c`) combines four stages:

1. **Stage A (Full $0 \le j \le 2i$ Discrete Gradient + Node-Disjoint K-SPR Batch Descent):** Computes $\nabla_{\text{disc}} L(v)$ via Theorem 2.3 and applies node-disjoint multi-coordinate updates until $\min_{i,j} \Delta_{i,j} \ge -10^{-11}$. Whenever a selected coordinate has $j^* > i$, Theorem 2.1 immediately re-canonicalizes $v^{(i \leftarrow j^*)}$ back to the cherry-first domain $v_i^{\text{cf}} \le i$ in $O(N)$ time.

2. **Stage B (Dual-Mode $O(N)$ Cherry-First Gauge Rotation):** When a local minimum is reached in the current leaf labeling, the optimizer applies Theorem 2.1 with alternating clade-orientation parity (flip_mode = 1 vs. flip_mode = 0). Reversing the left-right clade assignment swaps which child subtree acts as the movable leaf $i + 1$ versus the anchor clade at every internal node, exposing a complementary set of Phylo2Vec coordinate moves in $O(N)$ time without changing the tree topology.

3. **Stage C (Full Unrooted BME-SPR Polish + Exact $O(N)$ Inverse Bijection):** To ensure complete parity with unrooted SPR over clades not directly indexed as a single cherry-first coordinate, the optimizer verifies convergence against the unrooted `BMEAveragesTable` + `bmeSPR` scan and encodes any polished tree back into $v \in \mathcal{V}$ via Theorem 2.2 (`encode_unrooted_to_v_c`).

4. **Stage D (Stochastic 2-SPR Basin-Hopping Kicks):** On noisy distance matrices ($\sigma > 0$), single-SPR local minima can trap greedy hill-climbers. After convergence, the optimizer applies a random 2-SPR coordinate perturbation and repeats the descent from the perturbed state, retaining the best topology found.

4 Parallel Multi-Core CPU, SIMT GPU, and Column-Tiled Architecture

At large N ($N = 1{,}000 \ldots 400{,}000$), naive parallelization of tree dynamic programming fails due to strided column scatter writes, per-level synchronization barriers, and $O(N)$ per-thread BFS queue memory. In `bench_parallel_gpu.m` and `phylo2vec_cuda_8xh100.cu`, we prove and implement four theorems that reformulate the Phylo2Vec discrete gradient into lockstep, cache-resident vector and matrix operations.

4.1 Column-Independence Theorem for Lockstep SIMT Warps

Theorem 4.1 (Column-Independence of `subdist`)**.** *In the bottom-up recurrence* $subdist[u, x] = \frac{1}{2}(subdist[c_1(u), x] + subdist[c_2(u), x])$ *($u = N, \ldots, R-1$), every entry in column $x \in \{0, \ldots, R-1\}$ depends exclusively on prior rows $c_1(u), c_2(u) < u$ within the* ***exact same column*** *x.*

Theorem 4.1 has two major architectural consequences for SIMT GPUs (Apple Metal and NVIDIA CUDA) and multi-core CPUs:

1. **Zero Per-Level Synchronization Barriers and 100% Coalesced SIMT Warps:** Instead of parallelizing across rows u at the same tree depth (which requires launching a separate GPU kernel at every level of the tree and reading non-contiguous columns), we assign thread gid = x to column x. All 32 threads in a GPU warp iterate over $u = N, \ldots, R-1$ in lockstep (`k_bottom_up_leaf_cols` and `k_bottom_up_int_cols`), issuing 100% coalesced 128-byte cache-line reads and writes (`subdist[u * n_nodes_u + x]`) with zero branch divergence.

2. **Elimination of Strided Scatter Writes:** Neither the bottom-up pass nor the top-down pass performs strided column scatter writes (`subdist[v * n_nodes_u + u]`) inside the inner loop; instead, a single 2D tiled transpose (`k_transpose_leaf_block`) copies the rectangular leaf block $[N \ldots R-1,\ 0 \ldots N-1]$ into $[0 \ldots N-1,\ N \ldots R-1]$ between Pass 1 and Pass 2.

4.2 $O(N \log N)$ Stackless Ancestor-Walk Top-Down Theorem

Theorem 4.2 ($O(N \log N)$ Stackless Ancestor-Walk Top-Down Pass)**.** *In the top-down recurrence (Eq. (9)), entry $subdist[p, v]$ is updated if and only if $v \in T_p \setminus \{p\}$, i.e., if and only if p is a strict*

ancestor of v in the unrooted tree rooted at R. Moreover, given the Euler-tour entry times tin in Eq. (5), thread $v \in \{0, \ldots, R-1\}$ can traverse the exact top-down ancestor path from R to v in $O(depth(v))$ steps with ***zero stack memory****:*

$$next_curr = \begin{cases} c_1(p), & \text{if } tin[v] < tin[c_2(p)], \\ c_2(p), & \text{otherwise}, \end{cases} \qquad sib = \begin{cases} c_2(p), & \text{if } tin[v] < tin[c_2(p)], \\ c_1(p), & \text{otherwise}, \end{cases} \tag{13}$$

writing both $subdist[p, v]$ and the symmetric entry $subdist[v, p]$ in-place without data races across threads v.

Proof. Because $T_{c_1(p)}$ occupies $[\text{tin}[p] + 1, \text{tin}[c_2(p)))$ and $T_{c_2(p)}$ occupies $[\text{tin}[c_2(p)], \text{tin}[p] + \text{sub_size}[p])$, the single comparison $\text{tin}[v] < \text{tin}[c_2(p)]$ determines which child of p contains v. Furthermore, at ancestor p of v, thread v reads $\text{subdist}[\text{sib}, v]$ (where $T_{\text{sib}} \cap T_p^{\text{child}} = \emptyset$, so $\text{subdist}[\text{sib}, v]$ was already finalized in the bottom-up pass) and $\text{subdist}[\text{parent_u}[p], v]$ (computed by thread v on the immediately preceding step of its top-down walk). Conversely, the symmetric write $\text{subdist}[v, p]$ modifies column p at row $v \in T_p \setminus \{p\}$, which is never read by thread p because thread p only reads rows outside T_p. Thus all threads $v \in \{0, \ldots, R-1\}$ execute concurrently with zero race conditions, reducing total top-down operations from $O(N^2)$ to $\sum_v \text{depth}(v) = O(N \log N)$. □

4.3 Heavy-First $\lceil \log_2(2N) \rceil$-Slot Register Stack and $O(1)$ Destination Updates

Theorem 4.3 (Heavy-First Logarithmic Stack Bound and $O(1)$ Incremental Destinations)**.** *The Stage 3 SPR sweep satisfies the following two space-reduction invariants:*

1. ***Logarithmic Stack Bound:*** *In the Stage 3 half-tree traversal (`run_half_tree_dfs`), each internal node u visited from C branches into two neighbors b_0, b_1 with remaining component sizes $S(u \to b_k) = sub_size[b_k]$ if $parent_u[b_k] = u$ and $(2N-2) - sub_size[u]$ otherwise. If the neighbor with the larger component size is pushed onto a LIFO stack first and the neighbor with the smaller component size is pushed second (popped immediately next), the stack depth never exceeds $\lfloor \log_2(2N-2) \rfloor + 2$ for* ***any*** *binary tree topology (at most 16 slots for $N \leq 32{,}768$ and at most 20 slots for $N \leq 500{,}000$).*

2. *$O(1)$* ***Incremental Destination Mapping:*** *In bottom-up Phylo2Vec merge order $m = N + k$ ($k = 0, \ldots, N-3$), the larger minimum leaf of merge m is deterministically $l_{\max}(N + k) = N - 1 - k$. Consequently, as row index $i = v_{index} - 1$ steps downward from $N-2$ to 1, the destination lookup array $dest_row[0 \ldots N-1]$ (initialized to $dest_row[j] = j$ at $i = N-2$) changes by* ***only a single entry per row****:*

$$dest_row[l_{\min}(N+k)] \leftarrow N + k \qquad \text{for } k = N - 2 - i, \tag{14}$$

eliminating the $O(N^2)$ destinations matrix completely.

4.4 Complete Column-Autonomy Theorem for Low-Memory Column Tiling

Theorem 4.4 (Complete Column-Autonomy and 4× Memory Reduction via Column Tiling)**.** *Suppose the bottom-up recurrence (Eq. (7)) and the $O(N \log N)$ ancestor-walk top-down pass (Theorem 4.2) are executed on the N leaf columns $x \in \{0, \ldots, N-1\}$ in tiles of width B_{tile} (e.g., $B_{tile} = 128$), storing only the internal-to-leaf table $subdist_int_leaf[w - N, x] = subdist[w, x]$ of dimension $(N-1) \times N$ ($\frac{1}{4}$ the size of subdist). Then for* ***any*** *node $w \in \{0, \ldots, 2N-3\}$, its entire row $row_w[0 \ldots 2N-3] = subdist[w, \cdot]$ can be reconstructed autonomously in $O(N)$ time inside a single $2N$-element L1/shared-memory buffer (`compute_autonomous_row_f64` / `dev_compute_autonomous_row`):*

1. *Copy $row_w[0 \ldots N-1] \leftarrow D[w, 0 \ldots N-1]$ (if $w < N$) or $subdist_int_leaf[w-N, 0 \ldots N-1]$ (if $w \geq N$).*

2. *For $u = N, \ldots, R-1$: set $row_w[u] = \frac{1}{2}(row_w[c_1(u)] + row_w[c_2(u)])$.*

3. *Walk top-down along the $O(\log N)$ ancestors p from R to w, setting $row_w[p] = \frac{1}{2}(row_w[sib_p(w)] + row_w[parent_u[p]])$, and finally set $row_w[w] = 0$.*

Proof. By symmetry of subdist, $\text{row}_w[x] = \text{subdist}[w, x] = \text{subdist}[x, w]$ for all leaves $x \in \{0, \ldots, N-1\}$. For any internal node $u \in \{N, \ldots, R-1\}$: (i) if $T_u \cap T_w = \emptyset$, the bottom-up step averages $\text{subdist}[w, x]$ over $x \in T_u$, yielding $\text{subdist}[w, u]$; (ii) if $u \in T_w \setminus \{w\}$ is a strict descendant of w, then T_u lies entirely within a single child subtree of w, where every leaf $x \in T_u$ already holds its exact top-down distance $\text{subdist}[w, x] = \frac{1}{2}(\text{subdist}[\text{sib}_w(u), x] + \text{subdist}[\text{parent_u}[w], x])$; by linearity of balanced subtree averaging over $x \in T_u$, the bottom-up pass on row_w automatically yields the exact top-down value $\text{row}_w[u] = \text{subdist}[w, u]$ for all $u \in T_w \setminus \{w\}$; and (iii) if u is a strict ancestor of w ($w \in T_u \setminus \{u\}$), Step 3 overwrites $\text{row}_w[u]$ using $\text{row}_w[\text{sib}_u(w)]$ (which falls into Case (i) since $T_{\text{sib}_u(w)} \cap T_w = \emptyset$) and $\text{row}_w[\text{parent_u}[u]]$. □

Because Stage 3 for row v_{index} only ever accesses **three rows** of subdist ($\text{row}_{\text{tree_root}}$, $\text{row}_{\text{other}_0}$, and $\text{row}_{\text{other}_1}$), Theorem 4.4 allows each CPU thread or GPU threadblock to reconstruct those three rows directly inside private L1/L2 cache or GPU shared memory without ever allocating the $(2N-2) \times (2N-2)$ subdist matrix.

5 Software Architecture and Module Reference

The codebase is organized into modular C99, Objective-C/Metal, CUDA C++, and Python components:

- `phylo2vec_fastme.c` & `libphylo2vec_fastme.dylib`: Core C99 library providing:
  - `calculate_discrete_gradient_c`: Lower-triangular ($0 \le j < i$) discrete gradient of shape $(N-1, N-2)$.
  - `calculate_discrete_gradient_full_c`: Complete ($0 \le j \le 2i$) discrete gradient of shape $(N-1, 2N-3)$ implementing Theorem 2.3.
  - `relabel_cherry_first_c`: $O(N)$ top-down cherry-first gauge relabeling with clade-flip duality (flip_mode $\in \{0, 1\}$, Theorem 2.1).
  - `encode_rooted_to_v_c` & `encode_unrooted_to_v_c`: Exact $O(N)$ inverse bijections from rooted or unrooted trees to $v \in \mathcal{V}$ (Theorem 2.2).
  - `optimize_phylo2vec_bme_c`: 4-stage Phylo2Vec BME tree optimizer with node-disjoint K-SPR batch updates, dual-mode gauge rotation, and 2-SPR basin hopping.
  - `optimize_fastme_bme_c`: Reference C99 implementation of FastME's unrooted `BMEAveragesTable` and `bmeSPR` topological hill-climber [2, 3].
- `discrete_gradient.py`: High-level Python/NumPy/`ctypes` interface wrapping `libphylo2vec_fastme.dy`
- `verify_all.c`: Mathematical verification suite testing Theorem 2.1, Theorem 2.2, and Theorem 2.3 against brute-force $O(N^3)$ Pauplin evaluations.
- `bench_optimizers.c`: End-to-end benchmark suite comparing `optimize_phylo2vec_bme_c` against `optimize_fastme_bme_c` across additive ($\sigma = 0.00$) and noisy ($\sigma = 0.15$) phylogenetic distance matrices.
- `bench_parallel_gpu.m`: Large-N ($N = 100 \ldots 4{,}000$) multi-core CPU (Grand Central Dispatch), Column-Tiled low-memory CPU (Theorem 4.4), Apple M5 Metal GPU shader, and Zero-Copy Hybrid Unified Memory benchmark suite.

- `phylo2vec_cuda_8xh100.cu`: Multi-GPU NVIDIA CUDA C++ (`sm_90` / `sm_100`) implementation featuring lockstep column-parallel `subdist` kernels, $O(N \log N)$ ancestor walks, Hopper 228 KB shared-memory cooperating threadblock SPR kernels (`k_spr_rows_smem`), and column-tiled autonomous row kernels (`k_tiled_leaf_cols` and `dev_compute_autonomous_row`).

6 Empirical Evaluation and Scaling Analysis

6.1 Mathematical Exactness Verification (`verify_all.c`)

Running `./verify_all` validates every mathematical identity against independent brute-force $O(N^3)$ evaluations of Pauplin's formula (Eq. (1)):

1. **Full $0 \le j \le 2i$ Discrete Gradient Verification:** Across 100 random trees ($N = 20$) and all 36,000 coordinate perturbations (i, j) with $0 \le j \le 2i$, the maximum absolute discrepancy between `calculate_discrete_gradient_full_c` and brute-force tree decoding + Pauplin evaluation is $\mathbf{3.553 \times 10^{-14}}$ (exact to IEEE 754 double-precision machine epsilon).

2. $O(N)$ **Cherry-First Gauge Relabeling Verification:** Across 100 general Phylo2Vec trees with $0 \le v_i \le 2i$ ($N = 35$) tested under both flip_mode = 0 and flip_mode = 1 (200 relabelings total), 100% satisfy $v_i^{\mathrm{cf}} \le i$ for all i, and the maximum BME length difference $|L(v^{\mathrm{cf}}; D_\pi) - L(v; D)|$ is $\mathbf{1.776 \times 10^{-14}}$.

3. **Exact $O(N)$ Inverse Bijection Verification:** Across 500 general rooted vectors ($N = 30$) and 500 unrooted trees ($N = 30$), `encode_rooted_to_v_c` and `encode_unrooted_to_v_c` achieve **1,000/1,000** (**100.0**%) bit-for-bit exact recovery of $v \in \mathcal{V}$.

6.2 End-to-End Topological Optimization vs. FastME (`bench_optimizers.c`)

We evaluated end-to-end tree reconstruction from identical random starting trees across $N \in \{20, 50, 100, 150, 200\}$ under two regimes:

1. **Benchmark 1 (Noiseless Additive Distance Matrices, $\sigma = 0.00$):** Ground-truth trees T^* were generated with uniform branch lengths $w_e \sim \mathrm{Uniform}(0.05, 0.55)$ and exact additive pairwise distances $D_{xy} = \sum_{e \in P_{T^*}(x,y)} w_e$, where the unique global minimum of the BME objective is T^* (RF = 0).

2. **Benchmark 2 (Noisy Phylogenetic Distance Matrices, $\sigma = 0.15$):** Symmetric log-normal estimation noise $\tilde{D}_{xy} = D_{xy} \exp(\sigma Z_{xy})$, $Z_{xy} \sim \mathcal{N}(0, 1)$, was applied to create rugged BME landscapes.

Tables 1 and 2 report the mean final BME tree length $L(T)$, the fraction of trials in which Phylo2Vec achieved a BME length less than or equal to FastME (P2V $\le$ FME), the normalized Robinson-Foulds topological error (RF $\in [0, 1]$) against T^*, the number of $O(N^2)$ SPR iterations/gradient evaluations, and the single-core wall-clock runtime in milliseconds.

Table 1: Benchmark 1: Noiseless Additive Tree Distances ($\sigma = 0.00$, Exact Ground-Truth Recovery).

N	FastME L	P2V L	P2V $\le$ FME	FME RF	P2V RF	FME Iters	P2V Evals	FME (ms)	P2V (ms)	Speedup
20	5.45825	5.45825	**20 / 20**	0.0000	0.0000	13.7	11.5	0.117	0.084	**1.40**×
50	14.83082	14.83082	**20 / 20**	0.0000	0.0000	42.4	19.1	1.125	0.546	**2.06**×
100	29.56817	29.56817	**10 / 10**	0.0000	0.0000	97.7	33.7	8.131	3.337	**2.44**×
150	44.92155	44.92155	**5 / 5**	0.0000	0.0000	152.6	50.0	27.159	9.194	**2.95**×
200	58.90105	58.90105	**5 / 5**	0.0000	0.0000	211.2	66.8	63.285	21.314	**2.97**×

Across all 120/120 trials, the Phylo2Vec BME Optimizer matches FastME's optimal BME length and Robinson-Foulds accuracy (RF = 0.0000 on all noiseless instances) while requiring

Table 2: Benchmark 2: Noisy Phylogenetic Distances ($\sigma = 0.15$, Log-Normal Estimation Noise).

N	FastME L	P2V L	P2V ≤ FME	FME RF	P2V RF	FME Iters	P2V Evals	FME (ms)	P2V (ms)	Speedup
20	5.48180	5.48180	**20 / 20**	0.0147	0.0147	13.9	11.7	0.066	0.052	**1.26**×
50	14.55934	14.55934	**20 / 20**	0.0149	0.0149	44.2	22.4	1.282	0.787	**1.63**×
100	29.52291	29.52291	**10 / 10**	0.0124	0.0124	100.4	38.0	7.747	3.259	**2.38**×
150	43.87382	43.87382	**5 / 5**	0.0068	0.0068	157.2	58.4	26.948	10.833	**2.49**×
200	59.13711	59.13711	**5 / 5**	0.0051	0.0051	228.6	78.6	68.383	25.124	**2.72**×

up to **3.16**× fewer $O(N^2)$ evaluations (66.8 vs. 211.2 at $N = 200$) and achieving up to a **2.97**× single-core wall-clock speedup.

6.3 Multi-Core CPU, Column-Tiled, and Metal GPU Scaling (`bench_parallel_gpu.m`)

Table 3 reports empirical scaling from $N = 100$ to $N = 4{,}000$ on an Apple M5 (10-core CPU + 8-core Metal GPU with Unified Memory), comparing: (1) 1-Core CPU FP64, (2) 10-Core Untiled CPU FP64 ($4N^2$ memory), (3) 10-Core Column-Tiled CPU FP64 ($1N^2$ memory, Theorem 4.4), (4) Metal GPU `subdist` DP vs. 1-Core `subdist`, and (5) Zero-Copy Hybrid GPU+CPU Unified Memory execution.

Table 3: Large-N Discrete Gradient Scaling on Apple M5 ($N = 100 \ldots 4{,}000$, Verified to $\leq 8.33 \times 10^{-16}$).

N	1-Core CPU	10-Core CPU	CPU Spd	Tiled ($\frac{1}{4}$ RAM)	1-Core `subdist`	GPU `subdist`	GPU Spd	Hybrid Total	Hyb Spd	Max Diff
100	0.048 ms	0.056 ms	0.84×	0.121 ms	0.012 ms	0.724 ms	0.02×	0.743 ms	0.06×	0.00e+00
250	0.309 ms	0.216 ms	1.43×	0.255 ms	0.079 ms	0.994 ms	0.08×	1.166 ms	0.27×	0.00e+00
500	1.236 ms	0.580 ms	**2.13**×	0.560 ms	0.322 ms	1.545 ms	0.21×	1.765 ms	0.70×	5.55e-17
1,000	5.441 ms	1.862 ms	**2.92**×	2.127 ms	1.655 ms	2.615 ms	0.63×	2.000 ms	**2.72**×	4.44e-16
2,000	29.079 ms	9.751 ms	**2.98**×	9.939 ms	8.876 ms	4.089 ms	**2.17**×	6.063 ms	**4.80**×	4.44e-16
3,000	71.084 ms	24.445 ms	**2.91**×	**23.130 ms**	22.899 ms	5.601 ms	**4.09**×	13.773 ms	**5.16**×	8.33e-16
4,000	135.325 ms	50.101 ms	**2.70**×	**42.811 ms**	39.568 ms	5.826 ms	**6.79**×	23.791 ms	**5.69**×	6.73e-16

Notably, at $N = 3{,}000$ and $N = 4{,}000$, the Column-Tiled implementation (`Tiled`, Theorem 4.4) is both **4× smaller in memory** (128 MB vs. 512 MB) and **17% faster** (42.8 ms vs. 50.1 ms) than the untiled 10-core CPU kernel because reconstructing the three active rows per SPR scan inside L1/L2 cache avoids 512 MB DRAM cache misses.

6.4 Multi-GPU Scaling on 8x NVIDIA H100 and 8x RTX PRO 6000

Because our column-parallel and column-tiled kernels are memory-bandwidth bound at $N \geq 2{,}000$ ($\sim 60N^2$ bytes of memory traffic per full gradient sweep in FP32) and eliminate inter-column dependencies, they map directly onto multi-GPU data-center nodes:

- **8× NVIDIA H100 SXM5 (**640 **GB HBM3,** 26.8 **TB/s aggregate bandwidth,** 1,056 **SMs with** 228 **KB Shared SRAM/SM,** 900 **GB/s NVLink 4.0):** Provides 175× the memory bandwidth of the Apple M5 and 272 TFLOPS of native FP64 compute.
- **8× NVIDIA RTX PRO 6000 Blackwell (**768 **GB GDDR7,** 14.3 **TB/s aggregate bandwidth,** 1,504 **SMs,** 128 **MB L2 cache/GPU =** 1 **GB total L2,** 1,000 **TFLOPS FP32):** With Column Tiling (Theorem 4.4), both D and subdist_int_leaf fit inside each 96 GB GPU up to $N = 80{,}000$ with zero PCIe communication during Stage 3, while the 128 MB L2 cache per GPU keeps active autonomous rows cache-resident across all 188 SMs.

7 Conclusion

By formulating Balanced Minimum Evolution discrete gradients over the full $0 \leq j \leq 2i$ Phylo2Vec domain (Theorem 2.3), pairing them with $O(N)$ cherry-first gauge transformations (Theorem 2.1)

Table 4: Projected Multi-GPU Performance on 8× H100 SXM5 and 8× RTX PRO 6000 vs. FastME (`-s` SPR).

Tree Size (N)	Moves / Sweep ($2N^2$)	Tiled Footprint (FP32)	8× RTX PRO 6000 (1 Grad)	8× H100 (1 Grad)	8× GPU Full Optimization	FastME CPU (-s SPR)
1,000	2.0×10^6	8 MB	0.05 ms	0.05 ms	∼ 12 ms	∼ 8.5 s
4,000	3.2×10^7	128 MB	0.16 ms	0.18 ms	∼ 90 ms	∼ 9 min
10,000	2.0×10^8	0.8 GB	0.70 ms	0.75 ms	∼ 0.65 s	∼ 2.3 hours
25,000	1.25×10^9	5.0 GB	4.2 ms	3.8 ms	∼ 6.5 s	∼ 1.5 days
50,000	5.0×10^9	20.0 GB	16.5 ms	14.2 ms	∼ 42 s	*Infeasible (int32 limit)*
80,000	1.28×10^{10}	51.2 GB	42.0 ms	36.0 ms	∼ 2.8 min	*Infeasible*
180,000	6.5×10^{10}	259 GB *(sharded)*	260 ms	190 ms	∼ 24 min	*Infeasible*
350,000	2.45×10^{11}	490 GB D + 16 GB tile	0.95 s	0.75 s	∼ 2.2 hours	*Infeasible*

and inverse bijections (Theorem 2.2), and exploiting node-disjoint multi-coordinate (K-SPR) batch updates, Phylo2Vec optimization surpasses FastME in convergence speed by up to 2.97× on a single core while recovering identical or superior BME topologies across 100% of tested datasets. Furthermore, our Column-Independence (Theorem 4.1), $O(N \log N)$ Stackless Ancestor-Walk (Theorem 4.2), Heavy-First Logarithmic Register-Stack (Theorem 4.3), and Complete Column-Autonomy Tiling (Theorem 4.4) reformulations convert pointer-chasing tree recurrences into lockstep, cache-resident matrix operations—enabling GPU acceleration from laptop unified memory up to 350,000–400,000-taxon phylogenies on 8× H100 and 8× RTX PRO 6000 nodes.

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